

A Cryptographic Infrastructure for Friend-to-Friend Choreographies

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LAP 2026

23-27 September 2026, Dubrovnik, Croatia

A framework for

A framework for
friend-to-friend

A framework for
friend-to-friend
local-first

A framework for

friend-to-friend

local-first

coordination

A framework for

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and its (forthcoming) implementation

*To appear on
JLAMP 2026*

- Motivations to go friend-to-friend
(and not being "obscured by clouds")

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- ▶ Motivations to go friend-to-friend
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- ▶ A model for local-first coordination...
 - ▶ F2F-choreographies
 - ▶ Nearbybytes

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- ▶ Motivations to go friend-to-friend
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 - ▶ F2F-choreographies
 - ▶ Nearbybytes
- ▶ Cryptographic scaffolding

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Blessing and curse of clouds

Advantages of clouds

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- ▶ reduced costs for maintenance, facilities, personnel, software, and hardware
- ▶ anytime-anywhere motto
- ▶ dynamic reconfiguration (services can be replaced if needed)

No free lunch

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- ▶ Lack of transparency

Cloud-computing is inherently a "black-box"

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- ▶ Vendor lock-in

Switching to a different vendor can have prohibitive costs

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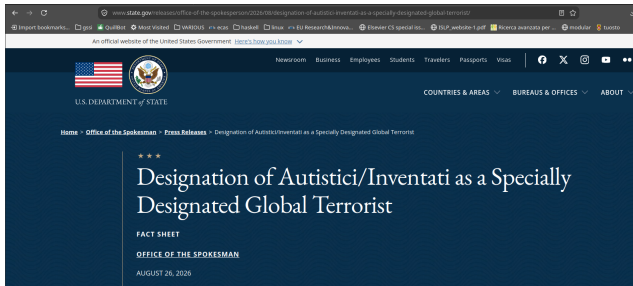
Switching to a different vendor can have prohibitive costs

- ▶ Dependence on connectivity

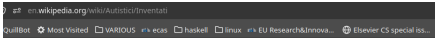
Cloud-based computing is heavily reliant on network connectivity

...anytime-anywhere almost always

...and limited autonomy/sovereignty



...and limited autonomy/sovereignty



Autistici/Inventati

Article Talk

From Wikipedia, the free encyclopedia

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Autistici/Inventati (AI) was an Italian **hacktivist** group that provided Internet communication services for groups opposed to **fascism**, **militarism**, **racism**, **sexism**, **homophobia** and **transphobia**.^[1]

Autist

Abbreviation	AI
Established	20
Dissolved	6 S
Type	Nc
Services	En

...and limited autonomy/sovereignty



Perhaps some re-thinking is needed?

en.wikipedia.org/wiki/Autistici/Inventati

as Monuments: Photograph a monument, help Wikipedia and win!

more

Autistici/Inventati

[Article](#) [Talk](#)

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"A/I" redirects here. For other uses, see [AI \(disambiguation\)](#).

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Autistici/Inventati (A/I) was an Italian **hacktivist** group that provided Internet communication services for groups opposed to **fascism**, **militarism**, **racism**, **sexism**, **homophobia** and **transphobia**.^[1]

Abbreviation	A/I
Established	2010s
Dissolved	6 September 2026
Type	Non-profit
Services	Encryption, web hosting, and other services

What if we...

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- ▶ kept reliance on the cloud to a minimum?
- ▶ retained control over
 - ▶ data (i.e., ensure confidentiality)
 - ▶ computational resources
- ▶ had more "autonomy"

Local-first coordination
via
Friend-to-Friend Choreography

Our choreographic model

Syntax

$$\begin{aligned} F &::= A \rightsquigarrow B : t \\ &\quad | F + F' \\ &\quad | F ; F' \\ &\quad | F \mid F' \end{aligned}$$

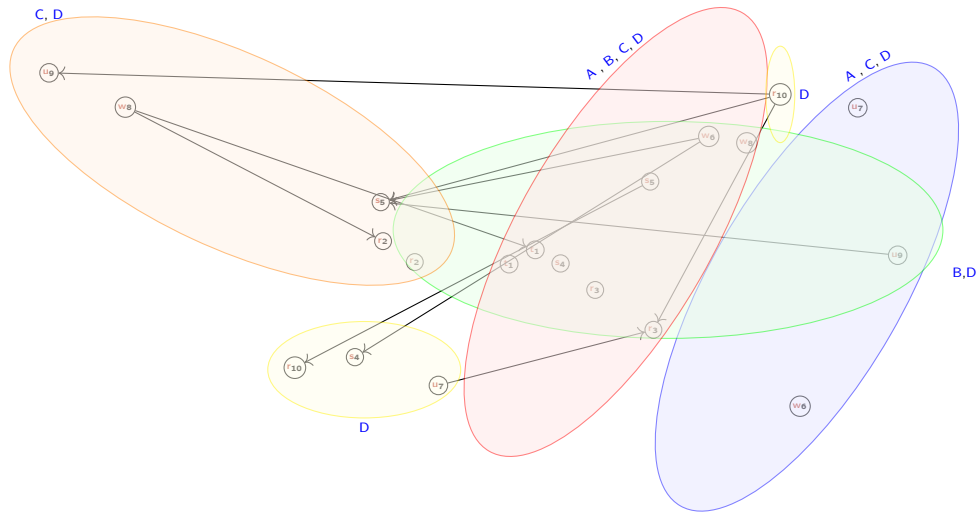
where

- ▶ B is a finite (possibly empty) set of A 's "buddies"
- ▶ t is a "type"

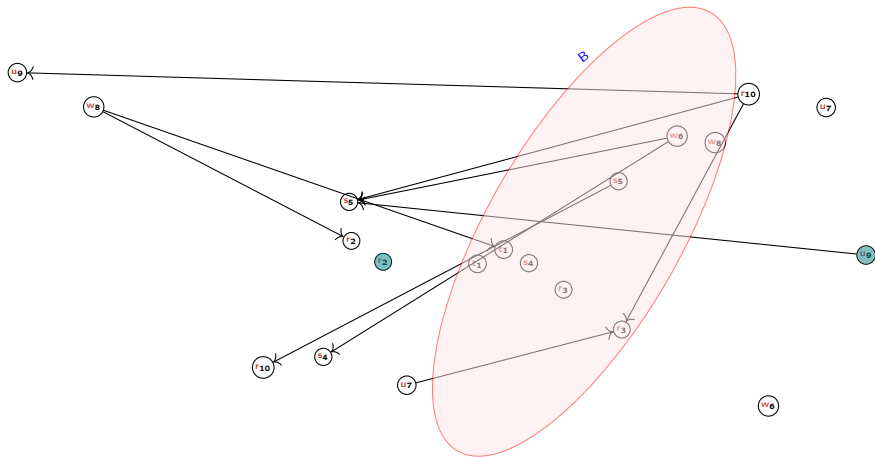
Semantics

$$\begin{aligned} \llbracket A \rightsquigarrow B : t \rrbracket &= \left\{ [A!B\langle t \rangle] \right\} \\ \llbracket F + F' \rrbracket &= \llbracket F \rrbracket \cup \llbracket F' \rrbracket \\ \llbracket F ; F' \rrbracket &= \{ r ;_{\varepsilon_r} r' \mid \dots \} \\ \llbracket F \mid F' \rrbracket &= \{ r ;_{\emptyset} r' \mid \dots \} \end{aligned}$$

The coordination model, informally



Replication



Having fun together

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A's options are between going out *for pizza* to a cinema or for a pizza at home

- ▶ if A opts for a cinema, B decides which movie; E should be oblivious of the title
- ▶ otherwise, deal is struck with whoever tells A which pizza they want

*since E would
complain anyway*

A and B share the

pizza

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Confidentiality constraints:

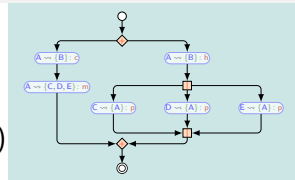
- ▶ E is not involved in the C-branch: can she join?
- ▶ X's choice of pizza is known only to X and A

An F2F-choreography for fun

$$\begin{aligned} F = & \quad A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ & + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p) \end{aligned}$$

An F2F-choreography for fun

$F =$ $A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m$
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An F2F-choreography for fun

$$F = \begin{array}{l} A \rightsquigarrow B : \textcolor{blue}{c}; B \rightsquigarrow \{A, \textcolor{blue}{C}, \textcolor{blue}{D}\} : \textcolor{blue}{m} \\ + \quad A \rightsquigarrow B : \textcolor{blue}{h}; (\textcolor{blue}{C} \rightsquigarrow A : \textcolor{blue}{p} \mid \textcolor{blue}{D} \rightsquigarrow A : \textcolor{blue}{p} \mid \textcolor{blue}{E} \rightsquigarrow A : \textcolor{blue}{p}) \end{array}$$

$$\llbracket F \rrbracket = \left\{ \left[\begin{array}{c} A!B\langle \textcolor{blue}{c} \rangle \\ \downarrow \\ B!\{A, \textcolor{blue}{C}, \textcolor{blue}{D}\}\langle \textcolor{blue}{m} \rangle \end{array} \right], \left[\begin{array}{c} A!B\langle \textcolor{blue}{h} \rangle \\ \swarrow \quad \downarrow \quad \searrow \\ \textcolor{blue}{C}!A\langle \textcolor{blue}{p} \rangle \quad \textcolor{blue}{D}!A\langle \textcolor{blue}{p} \rangle \quad \textcolor{blue}{E}!A\langle \textcolor{blue}{p} \rangle \end{array} \right] \right\}$$

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Let's spell out some requirements:

- ▶ h and c may be public
- ▶ how does E become aware that A decided to go out?
- ▶ how do C , D , and E know when A chooses to go out?

Nearbytes and its cryptographic infrastructure

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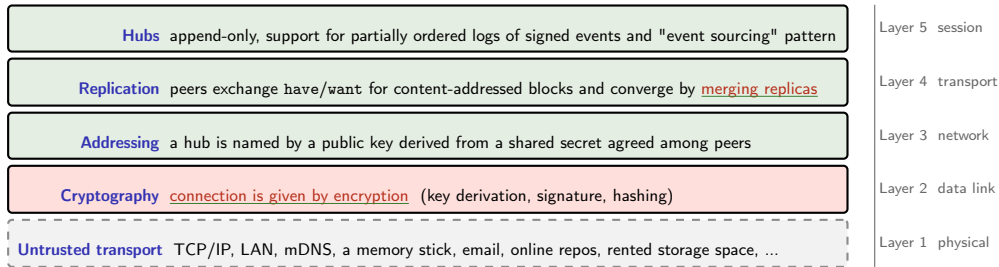
What is Nearbybytes?

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- ▶ set of protocols for confidential interactions combining
 - ▶ communication via "physical wiring"
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- ▶ logical content-based addressing based on public-key cryptography
- ▶ data-related events form a partially ordered log

Nearbytes's stack

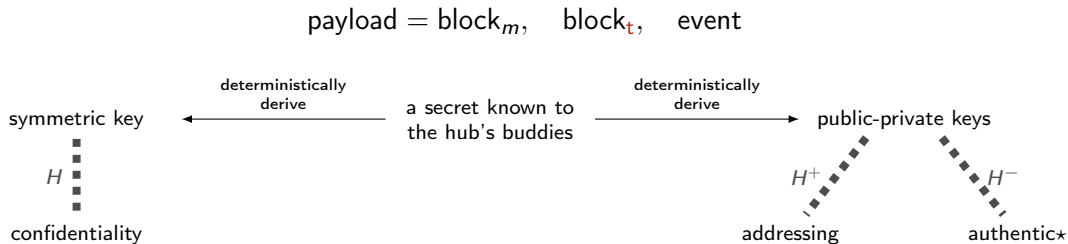


Nearbytes

Hush-hush (no key distribution)

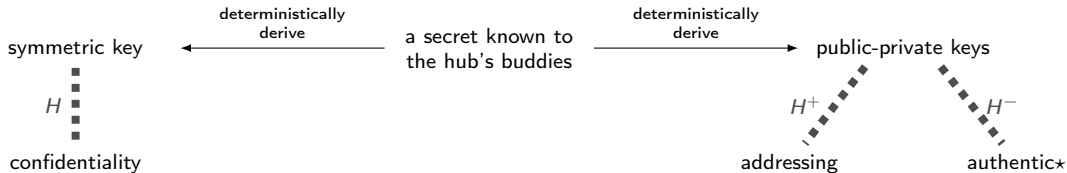
payload = block_m, block_t, event

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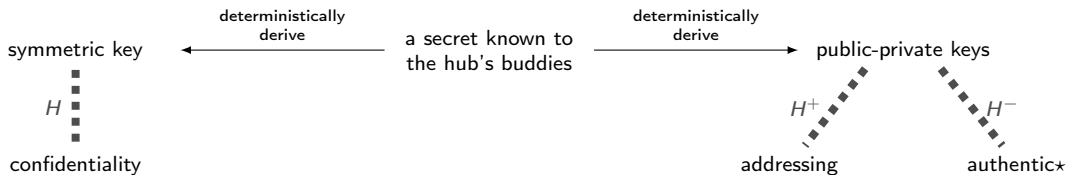
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$$\text{block}_m = \{\text{msg}\}_H$$

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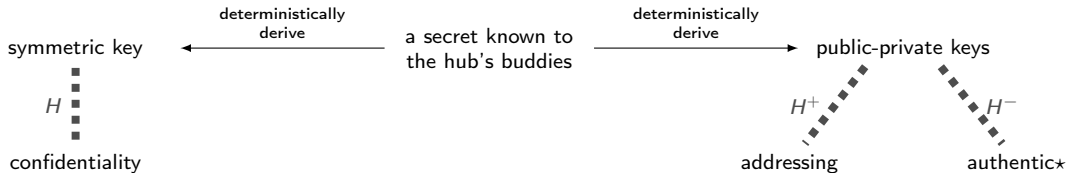
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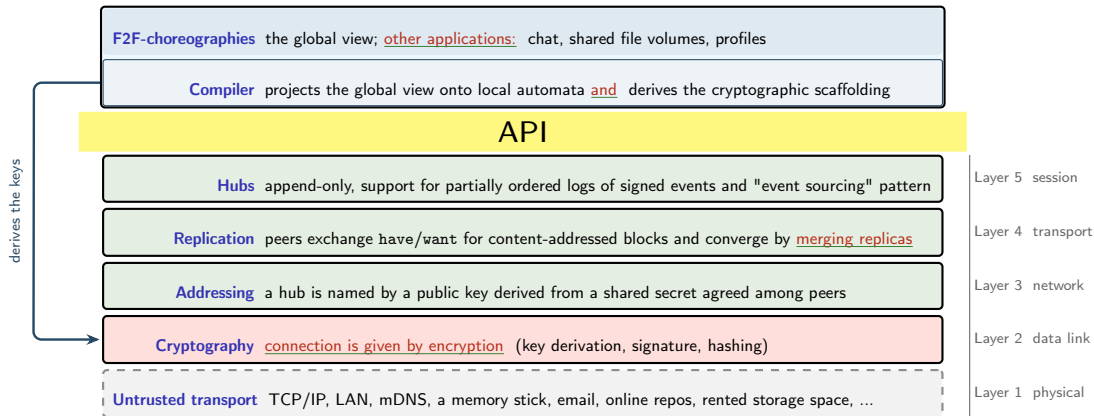
block_m = { msg }_H

block_t = { type }_{H⁺}

event = { hash(ev₁), ..., hash(ev_k), hash(block_m), hash(block_t) }_{H⁻}

F2F-choreographies & Nearbybytes

F2F-choreographies on top of Nearbytes



Hubs' addressing

Buddies agree on a "master secret" M

Leverage buddies' public keys

$$\left\{ \begin{array}{lcl} H & = & M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \cdot E^+ \\ H_1 & = & M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \\ \dots & & \\ H_{14} & = & M \cdot A^+ \cdot B^+ \\ H_{15} & = & M \end{array} \right.$$

Projections

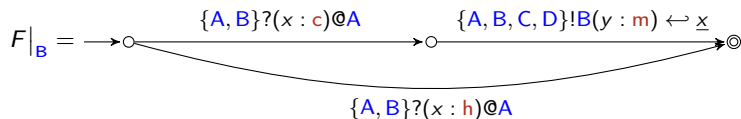
Deriving buddies' expected behaviour:

$$F = \quad A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p)$$

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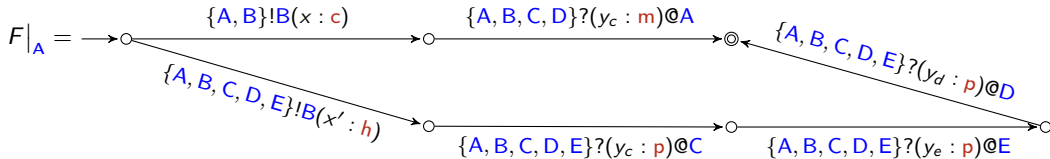
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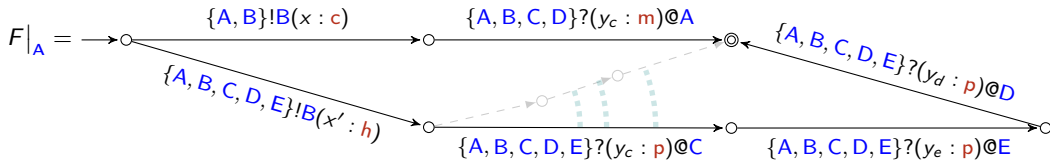
$$\left[A! \{A, B, C, D, E\} \langle h \rangle \right]$$

$$e = \{A \text{ 's address}\}_H, \{h\}_{H^+}, \{\text{hash}(\{A \text{ 's address}\}_H), \text{hash}(\{h\}_{H^+})\}_{H^-}$$

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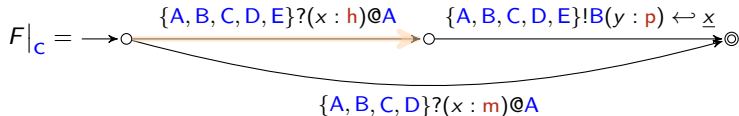
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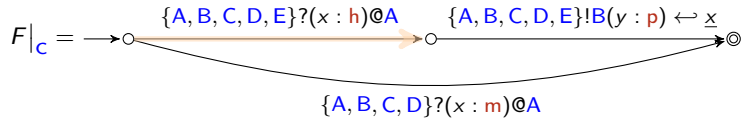
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$$\left[\begin{array}{c} A! \{A, B, C, D, E\} \langle \text{h} \rangle \\ \downarrow \\ C! \{A, B, C, D, E\} \langle \text{p} \rangle \end{array} \right]$$

$$e = \{A \text{ 's address}\}_H, \{\text{h}\}_{H^+}, \{\text{hash}(\{A \text{ 's address}\}_H), \text{hash}(\{\text{h}\}_{H^+})\}_{H^-}$$

$$\{\text{margherita}\}_H, \{\text{p}\}_{H^+}, \{\text{hash}(e), \text{hash}(\{\text{margherita}\}_H), \text{hash}(\{\text{p}\}_{H^+})\}_{H^-}$$

Thank you!

Obscured by clouds (Pink Floyd)

Bridges burning gladly
Merging with the shadows,
Flickering between the lines.
Stolen moments floating softly on the air,
Borne on wings of fire
and climbing higher.

Ancient bonds are breaking,
Moving on and changing sides.
Dreaming of a new day,
Cast aside the other way.

Magic visions stirring,
Kindled by and burning flames rise in her eyes.

The doorway stands ajar,
The walls that once were high.
Beyond the gilded cage,
Beyond the reach of ties.
The moment is at hand.

She breaks the golden band.